Journal of Building Engineering 98 (2024) 111050

Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe

Precision detection and identification method for apparent damage in timber components of historic buildings based on portable LiDAR equipment

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ARTICLE INFO

Keywords:
Historic timber buildings
Apparent damage
Detection
LIDAR
Non-destructive examination

ABSTRACT

For the purpose of preservation and restoration of historic timber buildings, it is critical that damage to timber components must be detected and identified accurately since proper detection and identification lay the foundation for scientific utilization and preservation of cultural heritage. Portable smart devices with integrated LiDAR sensor technology have significantly advanced, enabling them to be major pedestals for damage detection. An iPhone Pro based technique is therefore suggested for the precise detection and diagnosis of apparent damage to the timber components. The study consists of three major procedures. First, an apparent data collection of the timber components was conducted. A high-resolution triangular mesh model was utilized for precision detection and identification. Point cloud and texture data were the main constituents of this data collection. Second, a technique for image processing and apparent damage analysis was proposed for crack detection in historic timber buildings. This approach can be used to identify, recognize,and assess damage to timber components such as cracks, rotting, and knots. Finally, actual measurements of the timber components in Datong, Shanxi Province, at the Great Temple Corner, proved the effectiveness of the iPhone 12 Pro as a substitute approach to the conventional inspection equipment. The findings indicate that the iPhone 12 Pro, which has a built-in LiDAR sensor, performs exceptionally well in terms of effectiveness, accuracy, and graphical interpretation when identifying apparent damage to historic timber components. Moreover, it has the potential to substitute traditional measurement, 3D scanning, and other detection and identification techniques. Even though damage detection and identification of historical timber buildings has reached a mature stage in terms of portable, refined, and digital transformation, such technique can serve as a compatible addition as a safe, affordable, and paradigm shifting technical solution for detection and identification of historic building components.

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https://doi.org/10.1016/j.jobe.2024.111050
Received 29 October 2023; Received in revised form 25 September 2024; Accepted 11 October 2024
Available online 16 October 2024
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1. Introduction

China has a large number of historic timber buildings. In Beijing, there are more than 1700 cultural relics and monuments, with timber-frame historic buildings comprising more than 1200, which is approximately 71 % of the total [1]. Due to the age of construction, many historic buildings suffer from issues such as component damage. It is essential to explore more suitable restoration and strengthening techniques and reasonable utilization methods. However, a comprehensive record of the historical information and a proper research judgment of the damage to the components are necessary to achieve the least interference with the heritage and the maximum presentation and utilization of each value. The preservation and utilization could be achieved only by fully recognizing the value of the surviving heritage buildings.

At present, the apparent damage of historic timber components is usually detected and inferred by observation and topical measurement. As one of the most common types of damage to timber components in historic buildings, cracking affects the long-term service performance of timber components and is an important part of the stability and safety assessment of historic buildings. Due to the influence of environment, stress state and timber variability, precise detection of timber cracking is complex. There are difficulties in precise detection and restoration. For example, the existing technology is not comprehensive enough for detecting the apparent damage of timber components. Precision mapping as a fundamental work of heritage preservation, demands the records of existing state of historic buildings and apparent information in a comprehensive, complete and precise way, which can only be collected by full use of existing advanced scientific instruments and equipment. Besides, this supplies comprehensive and systematic basic information for further preservation of historic buildings [2]. Unlike traditional mapping and restoration surveys, “Precision Mapping” is to record specific geometric information or damage on heritage buildings [3]. Currently, the field of structural health monitoring of historic buildings is advancing in terms of methods and affordable costs. However, for the detection of specific damages on timber components, such as cracks and fissures, existing techniques have limitations, such as high risk, technical difficulties, awkward and expensive operations.

Advancements in LiDAR technology have enabled the new technical support of terrestrial and aerial laser scanning techniques (TLS and ALS) [4,5], and light drones [6] for detecting and identifying apparent damage in timber components. However, despite their effectiveness, high cost, operability, and safety concerns associated with these inspection systems make it challenging to be widely implemented for identifying damage in timber components.

The popularity of portable smart devices and advances in sensor technology offer the possibility of low-cost condition detection on historic building components. Contrary to traditional LiDAR systems, solid-state LiDAR sensors (SSL) embedded in smart devices allow for better scalability and reliability [7]. Besides, it has a longer lifetime and small-er size than the conventional LiDAR using a rotating mechanism [8]. Certain scholars have conducted studies in heritage recording, modeling, and detection. Murtiyoso et al. [7] verified the applicability of solid-state LiDAR sensors in portable devices for heritage recording, however, the accuracy results were strongly influenced by the scanning strategy, object apparent material, and point cloud density acquisition settings; Teppati et al. [9] from a cultural heritage perspective in The applicability of a miniaturized, low-cost portable solid-state LiDAR sensor was evaluated in different survey scenarios, and the analysis revealed that the quality of the acquired data depends on the size and characteristics of the measured object, acquisition time, and application algorithm; Balado et al. [10] compared three LiDAR systems, ground-based laser scanning, handheld mobile laser scanner, and iPad Pro LiDAR, for historical heritage masonry gates, verifying the characteristics of each technology in terms of inspection integrity, convenience and cost. In terms of building space inspection, Łabędź et al. [11] compared the results of data acquisition at different scales for sculptures, building interiors, and building facades using the LiDAR sensor on the Apple iPad Pro mobile device, and verified the applicability of the LiDAR sensor in the iPad Pro within 5 m. Mêda et al. [12]tested the use of LiDAR technology embedded in a mobile device for building renewal inspection. The use of iPad and BIMeo software allows for rapid data acquisition and analysis of the building interior space to quickly build a BIM model, but there are still shortcomings in the inspection process and procedures, etc. Li et al. [13] also explored the use of low-cost RGB D sensors to automatically generate interior as built building information models and achieved similar results.

Precise and accurate detection of crack geometry, skeleton curves, and development trends before restoration on timber components of historic buildings are essential in order to produce detailed data reports and image analysis results. Based on the existing research, a method of intelligent device with LiDAR sensor is proposed for expedient scanning, modeling and detection of timber components. Firstly, the timber components were subjected to surface data collection. A high-resolution triangular mesh model was used for accurate detection and identification. Secondly, an image processing and apparent damage analysis technique for crack detection in historic timber buildings was proposed to detect, identify and assess damage to timber components such as cracks, rot and scarring. Finally, a detailed data report and image analysis results were provided, and the effectiveness of this method was verified by actual measurements.

2. Materials and methods

2.1. Materials

The timber components used in this experimental study were sourced from residential buildings located in the Damiaojiao area of Datong, Shanxi Province. These included both dismantled, scattered timber components as well as those currently in use. The dismantled fragmentary timber components, mainly timber purlins, were taken from the abandoned residential timber frames in the area, which were built from the late Qing Dynasty to before the founding of the People’s Republic of China, and were mainly used for the measurement of the geometric characteristics and analysis of the cracking characteristics. The experimental timber was obtained from the site shown in Fig. 1.

2.2. Device and application

The portable LiDAR device used is the iPhone 12 Pro, released in October 2020, which has three RGB 12 MP rear cameras (wide angle, ultra-wide angle and zoom camera) and a LiDAR sensor with a total weight of 187 g and a built in bionic A14 processor (see Fig. 2). Compared to conventional LiDAR devices, the iPhone12Pro uses solid-state LiDAR (SSL), which eliminates the use of mechanical electric parts to enable superior scalability and reliability [7]., while offering a longer lifetime and smaller size than conventional LiDAR's that use a rotating mechanism [8]. The LiDAR sensor uses Sony CIS in combination with Lumentum's VCSEL and has a global matrix of 576 points, emitted in four different arrays of 144 points each; the global matrix consists of nine submatrices, each with eight points per column and eight points per row [14,15].The application used was Scaniverse version 2.1.2 from Toolbox AI to measure 3D apparent data of the timber components and generating color 3D models using LiDAR sensors. It relies not only on the embedded LiDAR sensors but also on other rear cameras to assist in the generation of textures and 3D models. Besides, with the portable device's own accelerometer, magnetometer, gyroscope and GNSS antenna, better hardware support is available for point cloud data acquisition and model building, an advantage not available with other types of LiDAR devices.

For the comparison verification section, different timber component testing equipment was used. Included are Apple's iPad Pro 2022, Revopoint's POP 3D scanner, and an accompanying motorized turntable. The Apple iPad Pro 2022 was released on October 18, 2022 and features an Apple M2 processor, a total weight of 466 g, a dual 12 MP+8 MP rear camera and solid-state LiDAR (SSL). scanning speed for clear texture recording. Built-in high-performance 3D computing chip with built-in projector and high-resolution IR camera, embedded 6-degree-of-freedom gyroscope, and 3D data scanning via high-resolution global exposure sensor, synchronized with depth camera via hardware trigger.

2.3. Device and application

The extent and form of damage on the components are complex owing to the complex structure of historical buildings, the variable form of the components, and the long history. This brings a lot of challenges to Precision mapping [2]. Existing technologies generally use stationary or handheld laser scanning equipment to first perform the acquisition of component point cloud data. This is followed by photogrammetric techniques to capture the apparent texture of the component. Afterwards, the data is transferred to a workstation where texture information is attached to the surface of the point cloud model to complete the processing and analysis of the raw data. The process is complex, arduous, and requires a high level of skill for the operator. Furthermore, usage of small sensors in architectural heritage preservation work is still in the exploratory stage. It requires hardware, software and theoretical support to reach the level of Precision mapping. Based on this current situation, a precision detection and identification method based on portable LiDAR equipment is proposed in order to comprehensively collect the damage traces and value information of the timber components. Besides, it is also to supply detailed inspection data and design basis for the restoration of historical timber buildings. The scanning strategy, data processing method, digital measurement and analysis method of this technology are studied, and validated in terms of efficiency and accuracy of inspection. The framework of this paper is shown in Fig. 3.

3. Apparent data collection and processing

3.1. Data collection

3.1.1. Data acquisition strategy

The effective detection distance of portable Lidar devices is generally within 5m, which is suitable for single component data collection at medium to close range. In the scanning strategy, the proposed spiral scanning method is adopted. Establish the main area of the damage, based on the starting point from the upper left side of the target from left to right spiral scanning. The specific data acquisition operation is: a whole timber component or part as the object to be measured, using the equipment to collect the point cloud and texture data of the object to be measured according to the established scanning strategy. The scanning trajectory is shown in Fig. 4.

Adopting this scanning strategy enables to effectively avoid the wide range of rotation of portable Lidar devices. This saves data acquisition time and reduces the jitter of the portable device during data acquisition. Therefore, improving the efficiency and accuracy of component apparent information acquisition.



Fig. 1. Experimental timber acquisition site - Damiaojiao area, Datong, Shanxi.



Fig. 2. Precision inspection method for apparent damage of timber components.

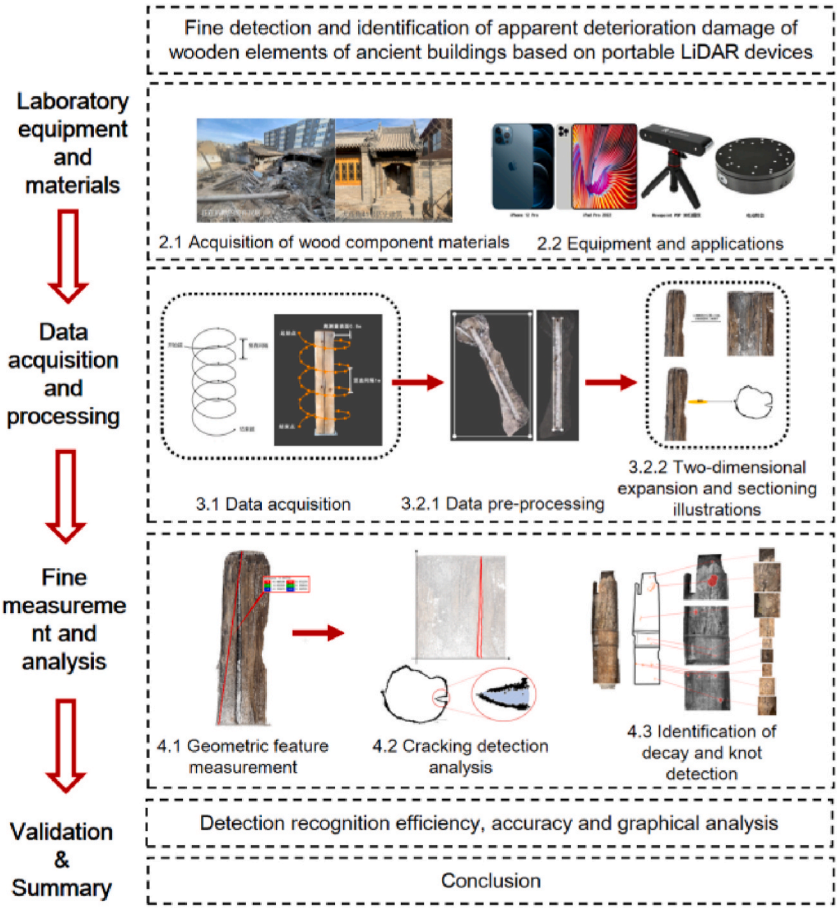


Fig. 3. Article framework.

3.1.2. Data acquisition strategy

The apparent damage inspection of timber components requires accurate inspection data and judgment on the current status of the damage, such as the length, width, depth and cracking trend of the timber components, the size of the rotten area, and the distribution of the location of the timber knot vortex, so the quality of data collection is demanding. When photographing, after locating the main cracks in the timber component, a spiral shot is taken along the square starting from the left side of the cracked end. The vertical interval is determined according to the focal length and shooting width to ensure that each layer could be shot with more than$ 30\% $ overlap. The attitude of the equipment is always kept within a certain range, and the vertical angle between the object to be measured does not exceed 30°, in order to minimize the impact of photographic errors on the closeup image.

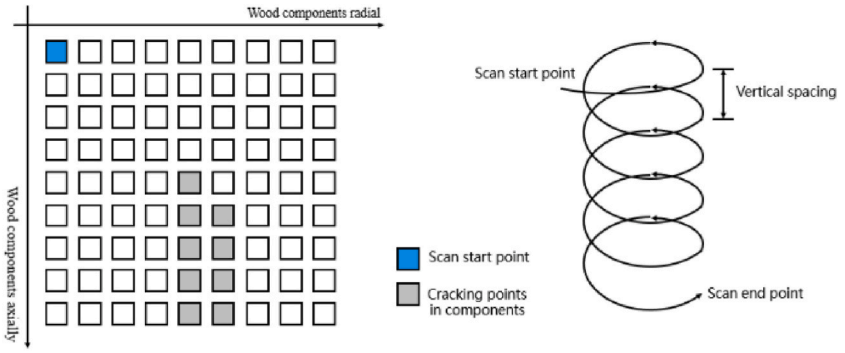


Fig. 4. Diagram of scanning strategy.

After the surround shot of the object to be measured, basic processing is carried out on the application side to get the rough model and raw data. Through inspection and comparison of the rough model, the basic features of the measured object and the environmental characteristics could basically be completely restored. The shooting process is shown in Fig. 5.

3.2. Data processing

3.2.1. Data image pre-processing

In the actual inspection environment, the timber components are often located in a complex environment. As shown in Fig. 6, it is extremely challenging to directly measure and analyze the rough mold and raw data. Based on the coarse mold, the timber components are identified, the target area is extracted from the coarse mold by rotating, scaling, and segmenting the coarse mold. After obtaining a single image of the timber component data, Precision detection and identification are then performed.

The point cloud data of the timber components is obtained after the rough model and the raw data are collated. In addition, it contains a large number of noise points caused by environmental factors, as well as some redundant data. To improve the accuracy of the model and to show the damage of the timber components more accurately and completely, the noise points are removed by the point cloud denoising function based on the removal of noninterest areas and obvious noise with the help of graphical display. In addition, the point cloud data need to be streamlined by keeping more data points in the areas of greater curvature and less data points in the areas of less curvature variation. The segmented point cloud data hereby allows the point cloud data to be streamlined using the resampling function [16]. As shown in Fig. 7.

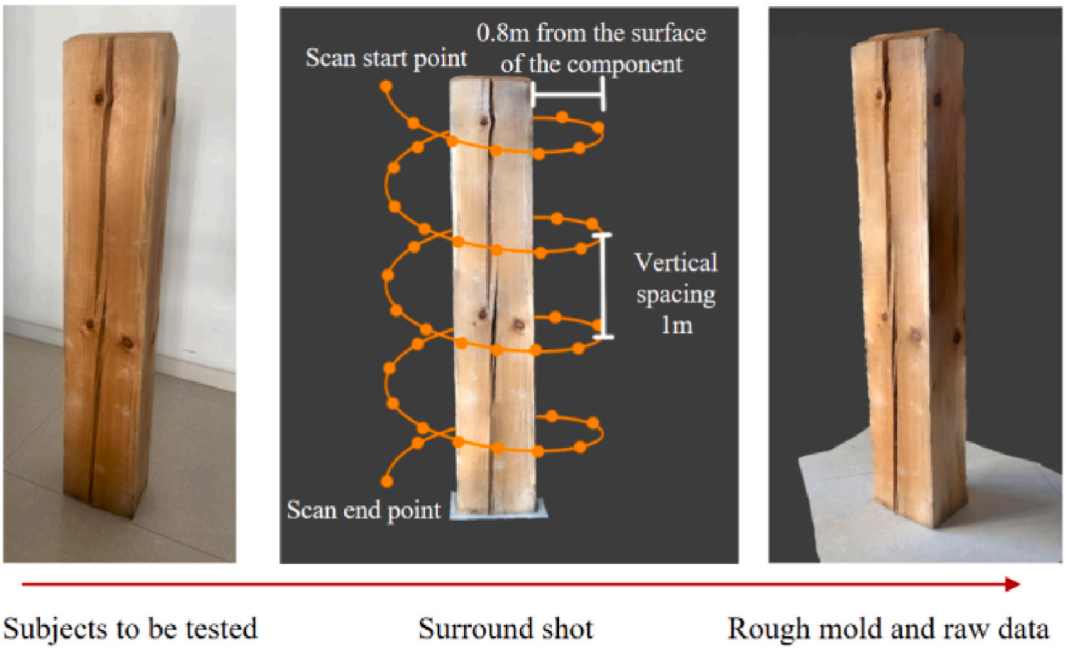


Fig. 5. Flowchart for filming and testing of timber components.

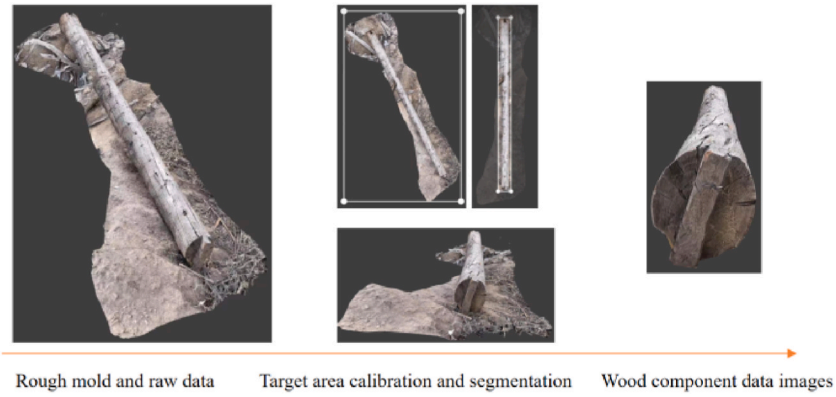


Fig. 6. Flow chart for calibration and segmentation of timber component areas.

3.2.2. Secondary processing of point cloud data

- Two-dimensional graphical interpretation method

The two-dimensional graphical solution method requires first determining the lengthwise and radial directions of the timber components. The data are projected to the two-dimensional plane according to the center of point cloud gravity in the radial section as an axis. The center of point cloud gravity is calculated using the center of gravity compression algorithm [17,18]: a rasterized center compression algorithm is used to keep the points closest to the center of gravity and remove the rest of the point cloud data. The original LiDAR point cloud is generally stored as a matrix after rasterization, and the minimum and maximum values of the *X,Y* and *Z* coordinates in the LiDAR point cloud data set are also obtained, thus forming a rectangular enclosing box parallel to the coordinate axes, which is used to envelop the entire LiDAR point cloud data. For LiDAR point cloud data $pi \in P$ to determine the size of the enclosing box, each point in *P* is searched to obtain the minimum and maximum $x_{\min}$ and $x_{\max}$, $y_{\min}$ and $y_{\max}$, $z_{\min}$ and $z_{\max}$, in the *X,Y* and *Z* directions, then the coordinates of the eight vertices of the enclosing box $\begin{matrix} x_{min} & y_{min} & z_{min} \\ x_{max} & y_{max} & z_{max} \end{matrix}$ these eight coordinates could determine the size of the enclosing box. The enclosing box is divided into a number of rectangular grids according to the demands of compression. Let the lengths of the 3 sides of the grids be d_x, d_y and d_z respectively, and according to the boundary of the enclosing box and the set size of the grids, the lengths of the 3 sides of the enclosing box parallel to the *X,Y* and *Z* axes are divided into *m, n* and *l* parts respectively, and the following relation is given:

$$\begin{aligned} a_i &= INT[(x_i - x_{min})/d_x] \\ b_i &= INT[(y_i - y_{min})/d_y] \\ c_i &= INT[(z_i - z_{min})/d_z] \end{aligned} \tag{1}$$

where a_i, b_i and c_i are the index numbers of the *X,Y* and *Z* axis directions of the raster where the point p_i is located, respectively, converted into a one-dimensional form as follows:

$$id = a_i + (b_i - 1)m + (c_i - 1)mn \tag{2}$$

Reorder the point cloud data in the enclosing box according to the index number *Id* in ascending order, then the point cloud data in the enclosing box is also stored consecutively according to the order of the raster, then calculate the number *N* of point clouds

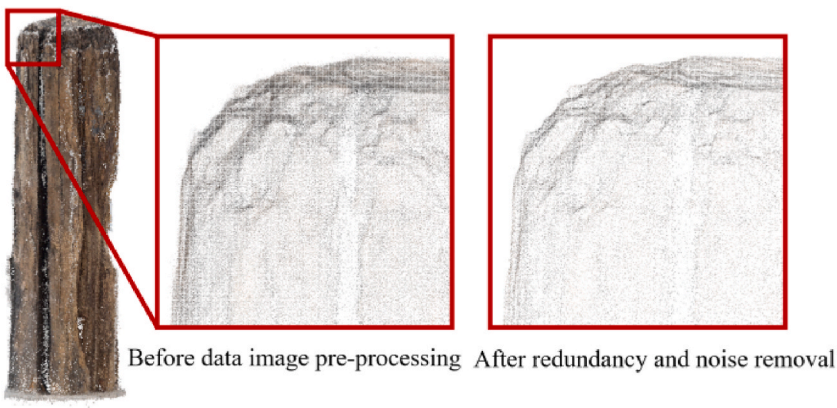


Fig. 7. Comparison of point cloud before and after denoising processing.

contained in each raster *Id*, let the coordinates of any point p_i in the raster be (x_i, y_i, z_i),calculate the position of the gravity center of all the point clouds in the raster (D_x, D_y, D_z):

$$D_x = \sum_{j=i+1}^{N} x_j D_y = \sum_{j=i+1}^{N} y_j D_z = \sum_{j=i+1}^{N} z_j \tag{3}$$

Calculate the distance from each point in the trellis lattice to the center of gravity d_i:

$$d_i = \sqrt{(x_i - D_x)^2 + (y_i - D_y)^2 + (z_i - D_z)^2} \tag{4}$$

Retain the nearest point to the center of gravity in the raster to obtain the center of gravity coordinates. By taking the gravity center of the point cloud as the axis and expanding it in two dimensions along the circumference of the section, a two-dimensional expansion diagram of the timber component could be obtained.

Based on this method, the reference plane given for this 2D illustration of the timber component is parallel to the internal base position. Fig. 8 shows a schematic diagram of the two-dimensional illustration along the axial midaxis. The two-dimensional illustration method could visualize the change and development of timber cracks on the plane, enabling powerful evidence for the next step of restoration and preservation research.

- Cutaway method

The sectioning illustration method allows for a detailed structural analysis of the timber components. Slices are created in both longitudinal and transverse directions (Fig. 9). The cross-sectional images at different heights are intercepted by setting the section spacing to analyze the changes indicated by the timber components (Fig. 10). The cross-sectional changes at different heights could be analyzed in the section images, including the change characteristics of decay and cracking. Besides, a comprehensive understanding of the structural characteristics on the timber components could supply digital data to support the next step of restoration.

4. Precision digital measurement and analysis

4.1. Measurement of the geometric characteristics

Timber components could suffer from damage such as decay and cracking after a long period of service. Traditional measurement methods are hardly able to accurately express the complex damage information. The shape of the component is not regular, so it is extremely hard to collect high precision dimensional information. However, with the help of Precision triangular mesh model, the required values could be directly obtained and the real geometric characteristics of the component could be obtained.

Based on the acquired high-precision point cloud data, a triangulated mesh model of a single timber component could be quickly constructed to measure the apparent damage information more clearly. Fig. 11 shows a model of five groups timber components modeled using the triangulated mesh framework, from a residential building being demolished in Datong, Shanxi.

The distance between points *A* and *B* in the triangular mesh could be measured by using the model to obtain the increments d_x, d_y and d_z on the *x, y* and *z* axes of the two points and the straightline distance between points *A* and *B*. This could be used to read the apparent base dimensional data of the components easily and quickly (Fig. 12).In the case of a timber post (Fig. 13), the upper and lower surfaces of the post are obscured from inspection when the post has not been removed from the building. Measuring the length of the post from the side using only a measuring tape will inevitably be affected by factors such as the inclination of the post and the curvature of the surface. Measurement through the model could circumvent the influence of these factors. After obtaining the coordinate data of the two points at the top and bottom of the timber post, the height between the two points obtained by incremental calculation is not affected by the posture and surface curvature of the timber post, and the results are more accurate.

The reliability of the precision triangulated network model measurements could be verified by measurement comparison (Table 1). The model measurement method could obtain finer measurement results. Based on the high-precision point cloud data and triangulation network model, accurate geometric feature data could be measured at smaller scales. According to the Specification for ancient

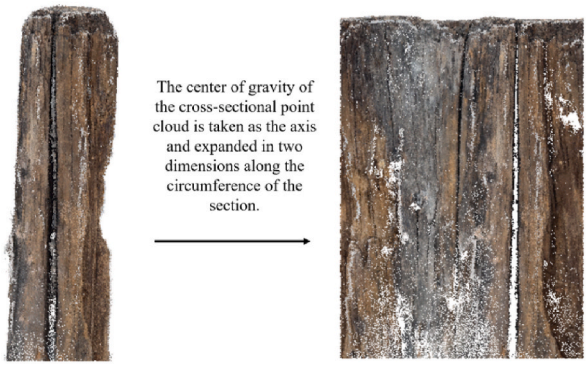


Fig. 8. Schematic diagram of a two-dimensional unfolded timber component.

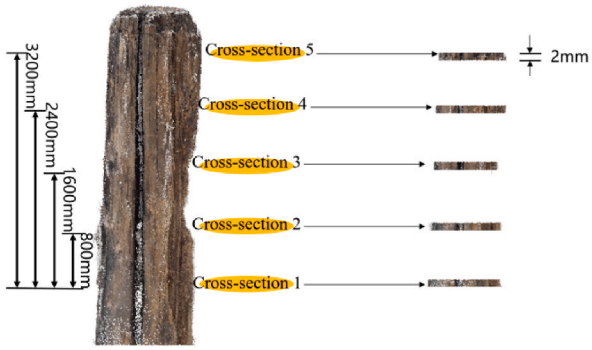


Fig. 9. Damaged timber posts with different height sections.



Fig. 10. Cross section of damaged timber column at different heights.

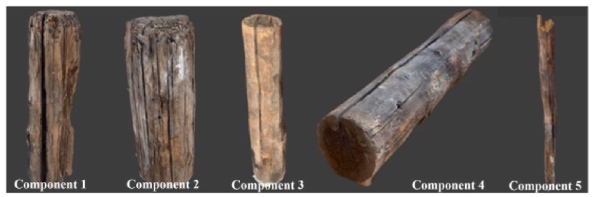


Fig. 11. Model of five groups timber components modeled.

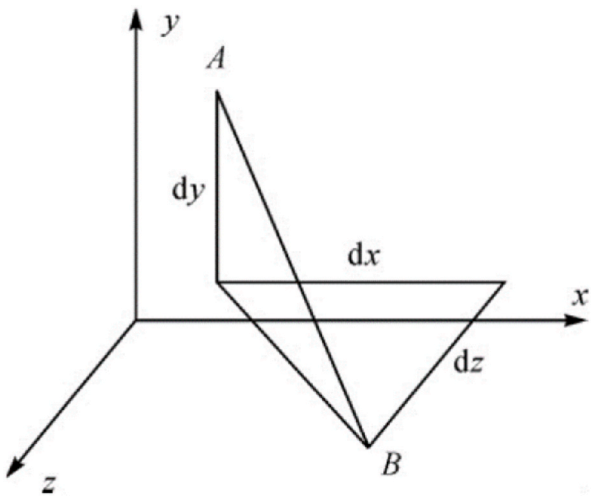


Fig. 12. Principle of dimensional measurement based on triangulation.

building surveying and mapping [19], the error limit in the size of the single building mapping, the error limit of the building detail size in the highest mapping accuracy class-I is 10 mm. After comparison, the difference between the model measurement and traditional measurement results are below the error limit in the specification, which could meet the measurement demands. Fig. 11 five groups of timber components measured information as shown in Table 1.



Fig. 13. Height measurement of timber components.

Table 1
Collation of information on the basis of five groups timber components.

Projects\ Component No.	1	2	3	4	5
Manual - Height (cm)	35.4	44.1	40.3	39.1	111.4
Model-height (cm)	34.92	43.71	39.65	38.77	110.93
Manual-radius (cm)	5.3	7	3.3	3.7	3.4
Model-radius (cm)	5.32	6.91	3.31	3.68	3.35
manual-Perimeter of member at 1/3 height (cm)	32.8	36.0	19.8	23.1	20.8
Model-perimeter of member at 1/3 height (cm)	32.96	36.36	20.13	23.24	21.25
manual-Perimeter of member at 2/3 height (cm)	32.4	36.0	19.8	23.0	20.8
Model-perimeter of member at 2/3 height (cm)	32.67	36.21	19.97	23.19	21.12
Number of points (k)	166	140	159	115	143
Number of triangular nets (k)	315	267	293	216	269

4.2. Analysis of cracked timber components

The curvature and discontinuity of the component surface could lead to incomplete judgement of cracking characteristics and development trends. To address this problem, the apparent shape and damage of timber components could be analyzed and studied using two-dimensional and sectional graphical methods.

4.2.1. Secondary processing of point cloud data

Take the example of cracking damage. The distribution of cracks on the timber component surface is clearly visible through the two-dimensional graphical illustration. Through the illustration (Fig. 14), it is found that on the whole outer surface of the timber component, starting from the top, there are two developing end cracks, two middle cracks in the middle, and one through-length crack running through the whole height of the timber component. According to the Technical Standards for Maintenance and Reinforcement of timber Structures in Historic Buildings, the Technical Specification for Safety Identification of Timber Components in Historic Buildings and the Code of Construction and Quality Acceptance for Construction Works of Historic Buildings, it is known that the 432.1 mm crack and the two 348.7 mm cracks are dangerous cracks; the 185.2 mm crack is a developing crack, and the crack development trend demands to be monitored; the two middle cracks are safety cracks.

4.2.2. Trend analysis of crack plane variation

The data on crack geometric characteristics such as crack location, plane size, height and width variation could be known from the two-dimensional illustrations. After the crack width was measured, a line graph was drawn for analysis (Figs. 15 and 16). The horizontal profile shows a trend of cracks widening, where the widest width appears at 80 mm height.

4.2.3. Analysis of the change trend in the cross-section

The trend of cracking and damage on timber components is visualized by measuring the depth of damage and cross-sectional area in the section diagram (Fig. 17). In traditional manual measurements, only the volume of timber components could be estimated. The cross-sectional illustration method allows comparison between intact and damaged timber components, and the difference between the areas is used to calculate an estimate size of the apparent damage area. In turn, the total volume of the internal cracked areas could be calculated, which helps to quantify the aid to the subsequent restoration work.

The variation of crack depth with height is reflected in Fig. 18, and the variation of cracking depth follows the same trend as the variation of cracking width. The cracking depth in the lower half is larger, and the maximum value appears near the height of 80 mm section. The difference between the maximum value and the minimum value is found to be more than 23.3 mm. In addition, according to the comparison of the development trend of crack depth and crack area, it could be seen that the crack is not uniformly developed within the crack, but the internal cracking space develops differently according to the different crack heights. The internal fracture size could be inferred from the relationship between the cracking depth and cracking area.

The changing trend of cross-sectional area, together with the data of crack width and depth, could supply an indepth understanding of the cracking situation and development trend. From Fig. 18, it could be seen that above 140 mm height, the cracking situation is not serious, and at 140–200 height, the cracking progresses rapidly and slightly slows down after reaching 80 mm height. In addition, it is also possible to quantify the damage to the appearance of timber components, such as the analysis of the change trend in the cross-sectional area of timber components (Fig. 19).

4.2.4. Analysis of crack development

After the statistical summary of the component cracking points in each section, the statistical results of its main cracking direction and cracking trend could also be obtained, and then the future cracking development could be predicted. From the statistical results, it could be seen that in addition to the main cracking in the 90° and 190° directions, there are small cracks in the 30°, 170°, 290° and 320° directions. Therefore, it demands to be paid attention to in the subsequent longterm monitoring. The two main cracking directions both have a northward and westward development (Fig. 20).

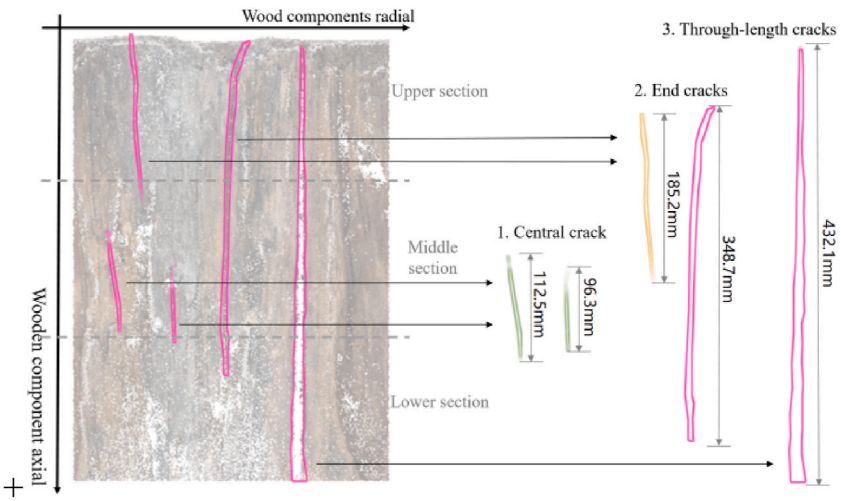


Fig. 14. Schematic diagram of crack shape location.

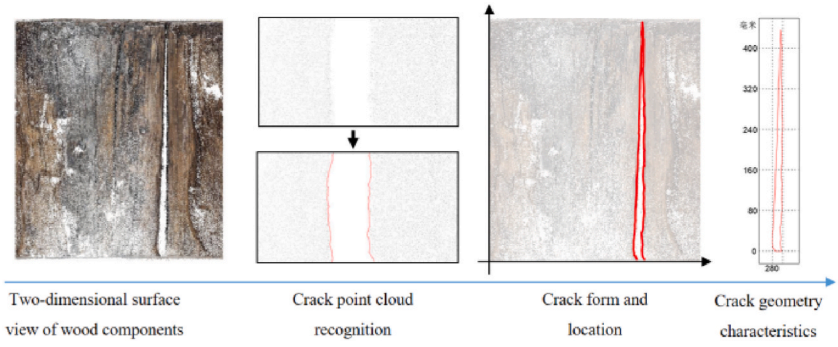


Fig. 15. Flow chart of crack geometry characterization.

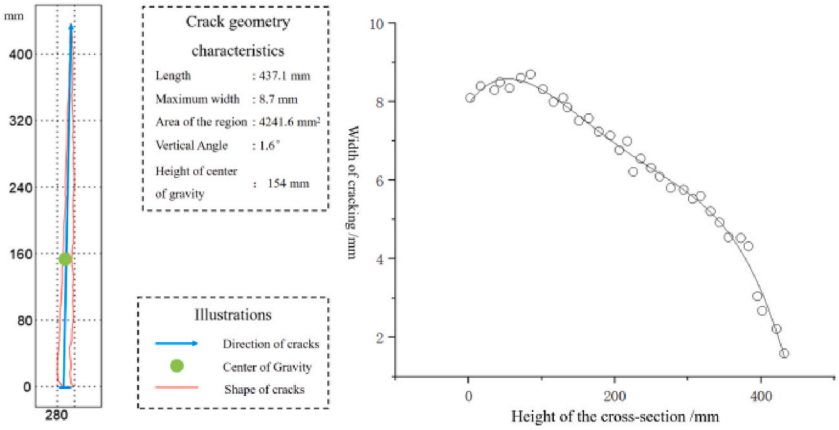


Fig. 16. Analysis of the crack's geometric features.

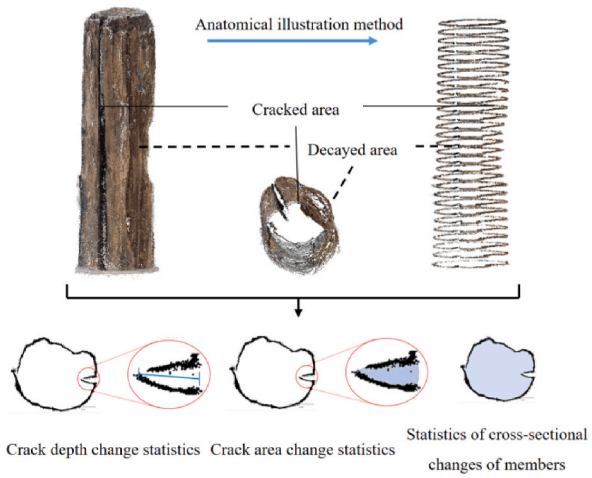


Fig. 17. Flow chart for spatial characterization of apparent damage to timber components.

4.3. Identification of timber decay and knot damage detection

After a long period of service and utilization, timber components are often subject to deterioration and decay, resulting in partial loss of the apparent timber. The traditional method of surveying timber components only records the size of decay and the distribution of knots, but lacks graphical representation and detailed data on such deterioration. By using graphical analysis, it is possible to accurately record the location and size of deterioration and knots on the surface of timber posts. For example, the three areas of deterioration shown in Fig. 21 below were identified by Precision inspection and the areas of deterioration were 85.04mm2,

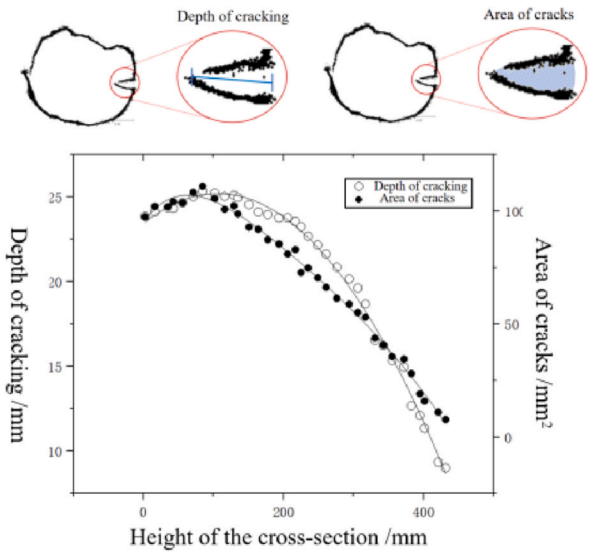


Fig. 18. Folding graph of change in cracking depth.

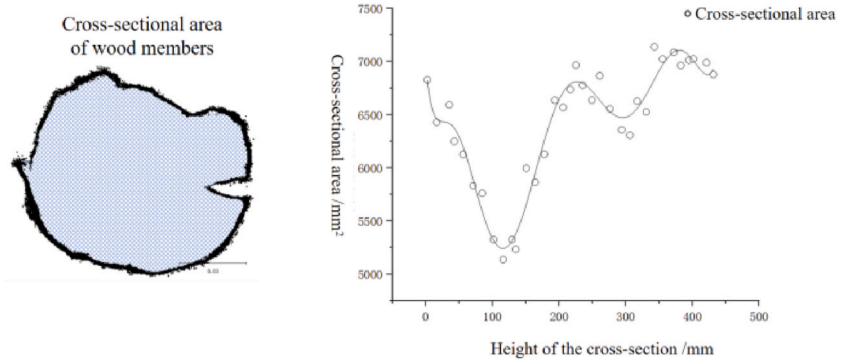


Fig. 19. Folding line diagram of the change in cross-sectional area.

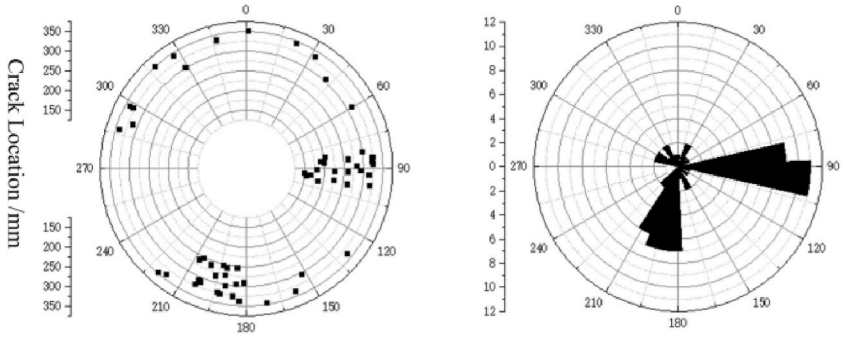


Fig. 20. Crack distribution and development analysis map.

172.7mm2 and 79.24mm2 respectively.

The natural defects in the timber components appearance could also be clearly documented, and these natural defects could have an impact on the mechanical properties of the timber components. As could be observed from the illustration, there are nine timber knots and swirls present in the timber column, three in the upper section of the old timber column and six in the lower end. This shows that the grade of material used for the restoration of the building at this location was not high and that there was no high demand for the selection of materials for the restoration of the timber components (Fig. 22).

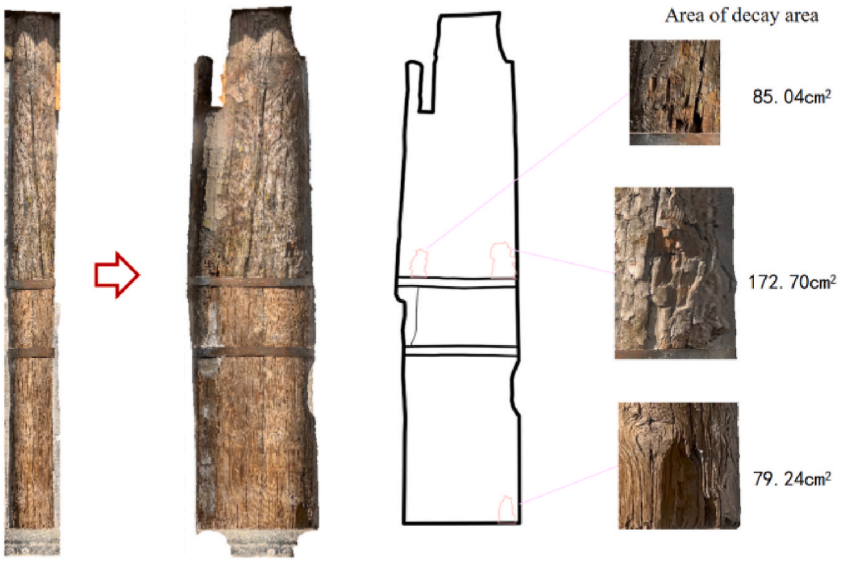


Fig. 21. Schematic diagram of the apparently deteriorated areas on a timber column.

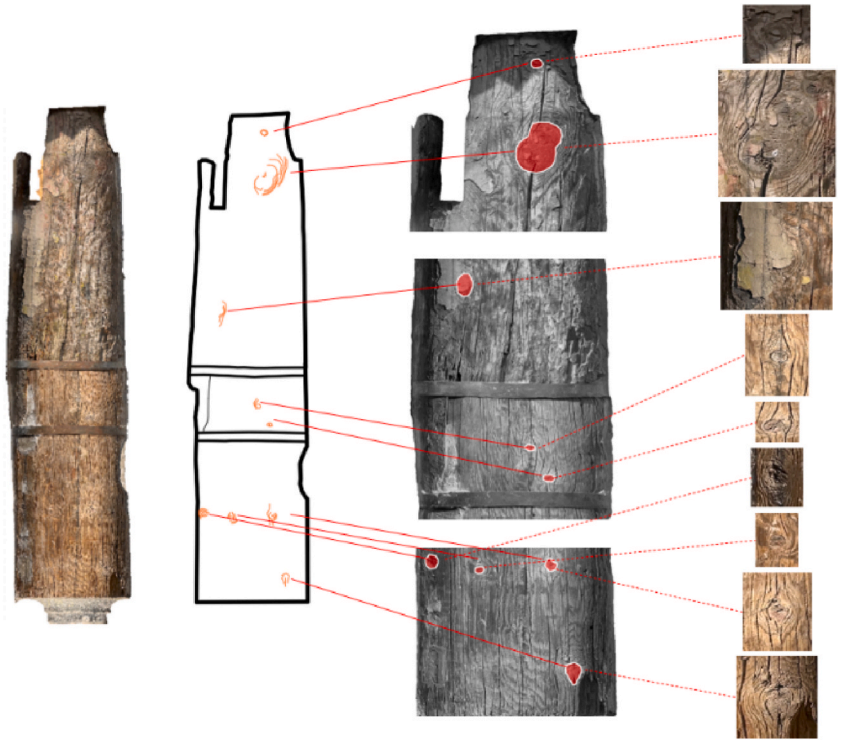


Fig. 22. Schematic diagram of the apparent knot swirl pattern on a timber column.

Convenient, longterm monitoring of the apparent damage according to the above method, observing the development of apparent damage, allows for efficient access to a richer monitoring material. This facilitates detailed interpretation of the apparent damage information on timber components.

5. Discussion

5.1. Comparison of damage detection and identification techniques

Currently, traditional manual inspection is mostly used to detect and identify the apparent damage characteristics of timber

components, and nondestructive testing instruments such as ultrasonic [20], stress wave [21], impedance meter [22], piro nail [23], and LiDAR scanner are partially introduced. The differences between portable LiDAR devices and currently available detection techniques are shown in Table 2.

Traditional manual inspection methods have certain drawbacks: low accuracy, restricted area, low amount of information, etc. It is challenging to make accurate judgment on cracking, damage characteristics and development trend. Nondestructive testing technology could not be applied to the recognition of the apparent damage detection at present. The characteristics of portable LiDAR equipment inspection methods are mainly reflected in the low cost, operability, and small potential hazards: (1) flexible and portable, easy to detect. Portable LiDAR data acquisition end using smart phones or tablet PCs, the size and weight of the device could support a single handheld. At the same time, it has proven to be meet the use of historic timber components in a small space for damage detection demands. (2) High integration. The data acquisition end module integrates a vertical surface emitting laser and a combined camera module, which could simultaneously complete laser scanning and photogrammetry, and could also assist in providing spatial depth, attitude angle, microdeformation, spectral information and other parameters of the apparent damage acquisition process data. (3) Arithmetic empowerment. Smartphone built-in high-performance processing chip, through the chip's CV algorithm support and application processing could be completed in a relatively short time after the data acquisition preprocessing.

5.2. Detection and identification efficiency

In order to verify the efficiency of this method in practical inspection work, several timber component surface inspection techniques were selected for comparative study. (1) portable LiDAR equipment, including Apple's iPhone 12 pro and iPad pro 2022. (2) 3D scanner equipment. 3D scanner equipment utilizes Revopoint's POP 3D scanner, which is available in two types: electric turntable fixed measurement and handheld measurement. (3) Traditional manual measurement, traditional manual measurement by the technical staff with many years of experience in historic building restoration project mapping in the field of simultaneous measurement. Through the same environmental conditions of multifrequency data collection statistical comparison analysis, statistics of its practical application in the prepreparation, measurement and scanning, data processing, quality testing, results output and other aspects of the comprehensive efficiency (Table 3).

The results in Table 3 show that.

(1) In terms of overall efficiency, portable LiDAR devices perform best, with the iPad pro2022 being able to perform automatic preprocessing faster due to the high performance of the built-in M2 processor, taking less time than the iPhone 12 pro. The handheld acquisition method of POP devices has a low fault tolerance rate, requires high measurement distance, speed and stability, and is prone to data alignment and modelling failure; motorized turntables Fixed measurement could effectively improve the success rate, but requires high environmental requirements and low overall efficiency. Traditional manual measurement takes the longest time and has the lowest overall efficiency.
(2) In terms of output, portable LiDAR equipment and 3D scanner equipment use the spatial and attribute information recorded in the point cloud, the amount of information in the results is richer, but the postindustrial processing requirements are higher; traditional manual measurement methods could quickly obtain simple poor loss data, the output results are relatively single, unable to form digital images.
(3) Workload, portable LiDAR equipment and handheld 3D scanner equipment external workload is 60 %–80 % of the traditional manual measurement workload, the internal work of data processing for external measurement to reduce a lot of workloads. At the same time, portable LiDAR equipment shows low technical threshold, high error tolerance, convenience and efficiency in field measurement, as well as the workload of external work is lower than that of 3D scanner equipment.

The main factors affecting the efficiency of portable LiDAR equipment are the familiarity of data acquisition methods, the complexity of data image processing and the degree of automation of data analysis. Therefore, optimizing the data acquisition method, simplifying and standardizing the data image processing process, and establishing an automated data image analysis method based on machine learning are the deepening research directions for the future Precision detection and recognition methods of the apparent components damage by portable LiDAR equipment.

5.3. Detection and recognition accuracy

With the growing focus on restoration and preservation of cultural relic buildings, there has been a gradual improvement in the standard of Precision mapping. The Technical standard for maintenance and strengthening of historic timber building [24], the Technical code on appraisal rating for wooden member of ancient timber building [25] and the Code for Construction and Acceptance of Construction in Traditional Chinese Architecture [26] give hazard grading and determination indicators for the apparent damage of timber components in historic buildings (Table 4). It is becoming increasingly important to accurately predict when timber structures will reach the hazard index and to carry out a higher standard of the apparent damage Precision mapping in order to arrive at a scientific determination of the hazard index of timber components.

Precision inspection for timber components has gradually gained attention, and the use of wearable devices [27], drones [28], machine learning [29], BIM [30] and other technologies to inspect timber components has largely enriched the methods and techniques available for inspection. However, for the more sophisticated cracks on timber components, existing studies rarely provide sophisticated measurements of their geometries, skeleton curves, and dimensional data, much less image-based inspection results. There is a lack of research results that allow for side-by-side comparisons. To assess the reliability and accuracy of the portable LiDAR-based method in the detection and identification of the apparent damage on historic timber components, the accuracy of the

Table 2
Comparative analysis of inspection techniques for historic timber buildings in China.

timber structure inspection technology	Mechanical properties of materials	Apparent residual damage detection	Internal defect detection	Damage caused by testing	Usage Cost	Operability	Accuracy of timber component inspection
Traditional manual measurement	×	✓	×	×	Low	Easy	Satisfy use
Resistance meter	✓	×	✓	√Micro loss	Medium	Medium Difficult	Complex impact
Stress wave detector	✓	×	✓	√Micro loss	Medium	Medium Difficult	Complex impact
Moisture content tester	✓	×	×	√Micro loss	Medium	Medium Difficult	High precision
Ultrasonic Detector	✓	×	✓	×	Medium	Medium Difficult	Complex impact
Ground Penetrating Radar	×	×	✓	×	Medium	Medium Difficult	Complex impact
Piro Nail	✓	×	✓	√Micro loss	Medium	Medium Difficult	Complex impact
Ground photogrammetry	×	✓	×	×	Low	Easy	Low precision
UAV photogrammetry	×	✓	×	× Risky	Medium High	Difficult	Low precision
Ground-based LiDAR	×	✓	×	×	Medium High	Difficult	Low precision
Aerospace LIDAR	×	✓	×	× Risky	High	Difficult	Low precision
Handheld LiDAR	×	✓	×	×	Medium High	Medium Difficult	High precision
Portable LiDAR devices	×	✓	×	×	Low	Easy	Satisfy use

Table 3
Efficiency statistics results.

No.	Work sequence	iPhone 12 Pro	iPad Pro 2022	POP Handheld	POP turntable	Traditional manual measurement
1	Preparatory work (measurement planning)	0.2h rough survey + 0.3h route planning	0.2h rough survey + 0.3h route planning	0.2h rough survey + 0.4h route planning	0.2h rough survey + 0.3h turntable arrangement	0.2h rough survey + 0.1h route planning
2	Scanning, measurement and survey operations	0.5h data acquisition	0.5h data acquisition	1.0h data acquisition	1.5h data acquisition	1.5h manual measurement
3	Field raw data processing	0.2h automatic pre-processing	0.1h automatic pre-processing	0.2h automatic pre-processing	0.2h automatic pre-processing	0.2h data review
4	Internal data post-processing	0.2h redundancy and noise removal + 0.5h secondary processing	0.2h redundancy and noise removal + 0.5h secondary processing	0.2h redundancy and noise removal + 0.5h secondary processing	0.2h redundancy and noise removal + 0.5h secondary processing	0.5h data logging collation
5	Internal data collation out of the map	1h bad loss calculation + 0.8h bad loss distribution	1h bad loss calculation + 0.8h bad loss distribution	1h bad loss calculation + 0.8h bad loss distribution	1h bad loss calculation + 0.8h bad loss distribution	3h bad loss distribution mapping
6	Quality Inspection	0.5h review check	0.5h review check	0.5h review check	0.5h review check	0.5h review check
7	Results collation	0.5h chart summary + 0.2h report formation	0.5h chart summary + 0.2h report formation	0.5h chart summary + 0.2h report formation	0.5h chart summary + 0.2h report formation	0.5h chart summary + 0.2h report formation
8	Total	4.9h	4.8h	5.5h	6.5h	7h (4h for bad loss data only)

data obtained by the portable LiDAR-based method was compared with that of the 3D scanner. Data were collected, processed and analyzed for five different groups of timber components using both fixed and handheld measurements on a motorized turntable. Five measurements were taken for each inspection method on a single timber component and the results were taken as the maximum value of the difference. Accuracy statistics are shown in Table 5 and Fig. 23.

The results in Table 5 and Fig. 21 show that.

(1) In terms of point cloud quality, the fixed position 3D scanner equipment with a motorized turntable for data acquisition obtains the highest number of component point clouds, as well as its point cloud data rate is higher and the operation is more efficient. However, the component must be able to be placed in a fixed position, which is more restricted by the operating environment. In contrast, portable LiDAR devices have a slightly lower number of point clouds than handheld 3D scanners, but have a higher success rate in realworld measurements. Portable LiDAR devices could obtain the same level of point cloud data as handheld 3D scanners.

Table 4
Determination of hazard indicators for timber components.

Category	Component type	Apparent damage phenomenon	Grading/processing requirements
Determination of the crack level of timber components in service	timber column	Signs of fracture, cleavage or crushing	Significantly affect or seriously affect the overall load-bearing capacity, should take immediate measures
	timber beam and square	Transverse interrupted cracking, cracks, or cracks are not seen, but the upper surface of the beam has signs of crinkle beam end splits (excluding dry shrinkage cracks) have end cracks or diagonal cracks caused by force or excessive deflection non-original sawing, slotting or drilling failed according to the remaining section test	
	Mortise and tenon	Cleaved or fractured	
	Short column、bracket、shaped suport	With splits	
	eave purlin、inverted V-shaped brace	Splintered or broken	
Determination of dry shrinkage cracks in timber with natural defects	–	Not allowed to appear on the shear surface of the connection	
		Near the shear surface of the joint, the depth of the crack (the sum of the two when there is an opposite crack) should not be greater than 1/2-1/4 of the diameter and 1/3-1/4 of the width of the material	
Determination of the grade of successive reinforced timber components	timber column	The column body has new deformation or dislocation, or the mortise and tenon has been cracked, or the iron hoop has been loosened	
		The original grout grout and timber bonding condition is poor, the grout dry shrinkage, knocking air drum sound; column body has obvious pressure wrinkle or deformation phenomenon	
		The original digging and patching area has been loosened, or new decay has occurred	
	timber beam and square	The original splice has been deformed or the bolt has been loosened	
		Dry shrinkage of the original grout, hollow drum sound when struck, or increased deflection of the beam body	

Table 5
Accuracy statistics.

Measurement items	timber components	iPhone 12 Pro	iPad Pro 2022	POP Handheld	POP turntable
Number of point clouds (K)	Component 1	166	170	199	266
	Component 2	140	142	171	215
	Component 3	159	162	218	254
	Component 4	115	114	149	167
	Component 5	143	145	178	221
Length (mm)	Component 1	349.2	349.2	349.1	349.3
	Component 2	437.1	437.0	437.1	437.1
	Component 3	396.5	396.4	396.7	396.4
	Component 4	387.7	387.5	387.6	387.6
	Component 5	1109.3	1109.3	1109.2	1109.3
Radius (mm)	Component 1	53.0	52.9	53.0	53.0
	Component 2	71.2	71.2	71.1	71.1
	Component 3	32.5	32.0	32.3	32.9
	Component 4	36.8	37.1	37.0	36.8
	Component 5	34.3	34.2	34.1	33.9
Crack width (mm)	Component 1	6.72	6.71	6.70	6.66
	Component 2	8.37	8.40	8.39	8.37
	Component 3	3.25	3.25	3.18	3.19
	Component 4	4.21	4.22	4.18	4.20
	Component 5	2.43	2.43	2.48	2.44

In addition, based on the high-resolution camera module, the portable LiDAR device could acquire point cloud data on the surface of the component and obtain high resolution images at the same time, which is an advantage that 3D scanners do not have. It has proven to be supply more intuitive and clear data support for the subsequent digital image analysis, and improve the efficiency and accuracy of data analysis.

(2) In terms of component data measurement, the data from the portable LiDAR device and other acquisition devices are compared in terms of absolute error and standard deviation respectively. The absolute error Δ is the measured value of the portable LiDAR

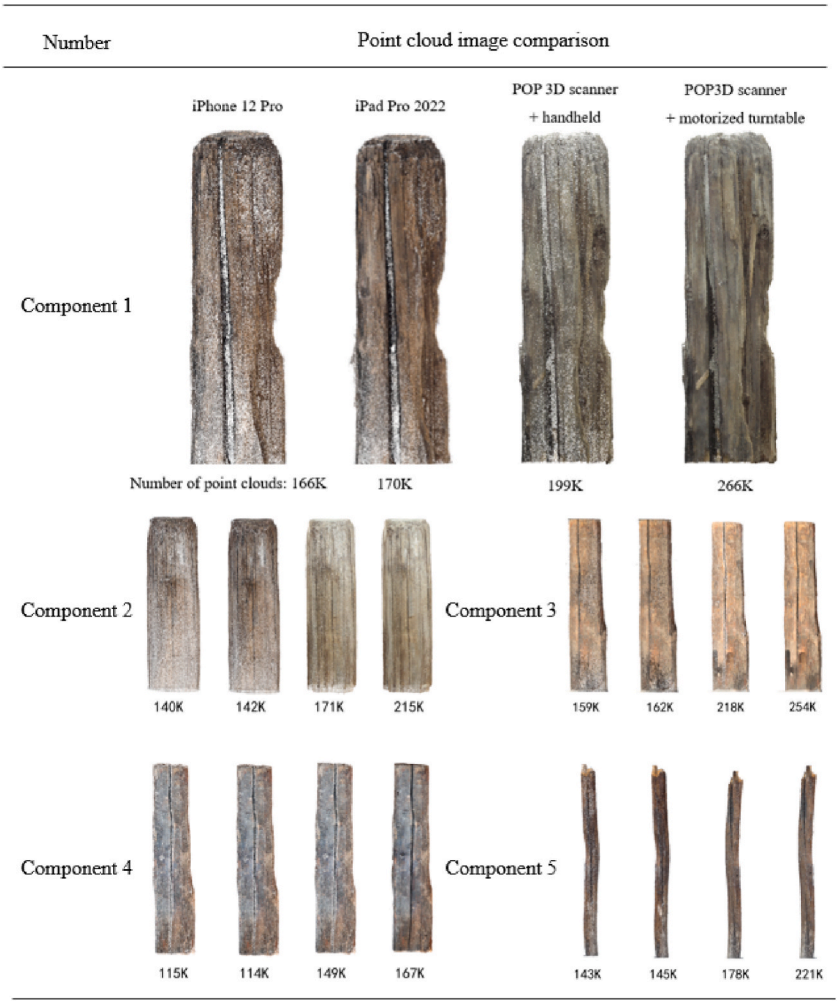


Fig. 23. Comparison of point cloud scan data images.

device minus the reference value of the data from other acquisition devices. The standard deviation is the square root of the median square of the set of true errors under the same observation conditions and is a numerical measure of the accuracy of the observation. The maximum Δ of component length is 0.2 mm, the maximum Δ of radius is 0.5 mm and the maximum Δ of crack width is 0.07 mm. The maximum standard deviation of component length is 0.095 mm, the maximum standard deviation of component radius is 0.075 mm and the maximum standard deviation of crack width is 0.033 mm. The performance of the portable LiDAR device in terms of accuracy and point cloud quality is basically the same as that of 3D scanning instruments.

In the data acquisition and model building process of portable LiDAR equipment, the main factors affecting the accuracy of the model are the number of point clouds and the number of triangulation networks. The point cloud density of the currently used portable LiDAR equipment is moderate and suitable for the collection of apparent damage information of timber components in general. The number of triangles depends on the accuracy of the construction of the point cloud data, and could be chosen to match the size and accuracy requirements of different volumes of components when generating the triangles. A more detailed triangulation model with sufficient detail retention requires longer data processing time and a certain amount of digital processing power of the equipment, with certain hardware requirements.

5.4. Formatting of mathematical components

Portable LiDAR equipment has the capability to retain amounts of digital images, which could be utilized for accurate identification and detection of damage caused by deterioration. This could aid in understanding the extent of damage to timber components through both internal and external measurements. A multi-dimensional analysis of the damage is generated to supply a reference for subsequent restoration, preservation and research analysis(Fig. 24).

In the precision detection and identification illustration of portable LiDAR equipment, the way in which the technician operates, the degree of automation in the identification of cracks and the crack pattern have a significant impact on the results. After obtaining

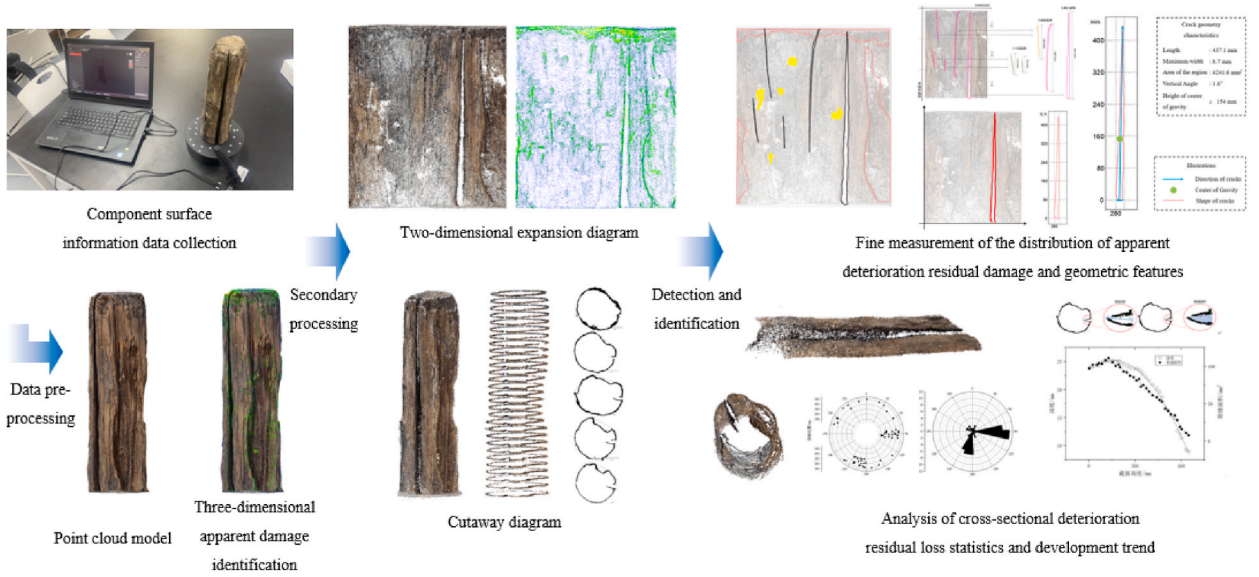


Fig. 24. Illustration of timber component apparent damage detection and identification.

the apparent data of the timber components, the two-dimensional unfolded and profiled sections allow for statistical analysis of the apparent damage on the timber components in different ways. After the 2D unfolding of the planes, a 2D plan view of the timber component could be obtained, on the basis of which the location of cracks, geometric features, etc. could be located. This process currently takes manual data extraction, which could be used to extract the required data as needed, but requires a certain level of operator competence; the graphical analysis based on cut sections has some depth data for the apparent decay and cracking of timber components, but for timber component cracking, only larger cracks could be obtained through portable LiDAR equipment to obtain internal crack data, which is less applicable to smaller cracks.

Table 6 shows that the detection capability of portable LiDAR equipment is the same as that of 3D scanners and traditional manual inspection in the detection of apparent damage on timber components. However, in terms of graphical analysis, portable LiDAR equipment could be used to digitally and graphically represent the crack distribution, crack geometry, crack development trends, damage distribution and area geometry of timber components, and support digital modelling, AR inlay and other multi-purpose open format exports to support indepth analysis on this basis, which is not possible with other inspection methods. difficult to achieve. Through the refinement of the portable LiDAR device, the full information components of the timber components could be collected and identified by the method of Precision mapping, so that the apparent damage information and historical microtraces of the timber components could be graphically expressed and numerically analyzed, realizing the intelligent expression of “Precision Mapping” at the level of timber components.

Table 6
Comparative analysis of inspection techniques for historic timber buildings in China.

Apparent appearance of timber components Damage test items	Portable LiDAR detection and identification method	Handheld 3D Scanner	Electric turntable + 3D scanner	Traditional manual testing
Component size measurement	✓	✓	✓	✓
Cracking detection	✓	✓	✓	✓
Crack distribution characteristics Graphical representation	✓	×	×	×
Crack geometry characteristics Graphical representation	✓	×	×	×
Cracking development trend Graphical analysis	✓	×	×	×
Decay, timber knots, etc. Damage detection	✓	✓	✓	✓
Graphical representation of the distribution characteristics of damage such as decay	✓	×	×	×
Precision measurement of deteriorated area features such as spoilage	✓	×	×	×
Identification of apparent micro-trace inspection of components	✓	×	×	✓
OBJ, PLY, STL base format export	✓	✓	✓	×
Digital modeling, AR inlays, and other multi-purpose open format standard exports	✓	×	×	×

6. Conclusion

Restoration and preservation of historic timber buildings is a systematic research work with rigorous logic and analysis. In addition to ensuring the physical solidity of the building, the key lies in preserving the integrity value of historic buildings and preventing irreducible damage to the original data or physical information. Portable LiDAR devices define a model for the detection and identification of damage to timber components. In this manner, smart phones or tablets serve as a platform for detailed mapping of historic buildings because of their affordability, operability, and versatility. It is more compatible for the detection and identification of damage caused by apparent degradation to timber components of historic buildings. With a small survey scale, it has proven to be effective in gathering timber apparent data with high efficiency and cheap cost and identifying deterioration damage with high clarity and accuracy. In order to supply complete and accurate digital basic information for the restoration design of historic buildings, as well as to safeguard and read the value information of the timber components at all stages, portable LiDAR devices can be a useful tool.

After thorough testing and analysis, iPhone 12 Pro has demonstrated exceptional performance in the field of detection and identification technology. It has proven to be highly efficient and accurate in detecting and recognizing apparent damage of historical timber components. As a result, it has the potential to replace existing tools used for this purpose. In particular, its support for common file formats and repository features used for historic timber building site data will greatly promote the establishment of a cloud data platform for damage to timber components. In turn, it will lead to public participation or upload through crowd testing of engineering restoration units to achieve data sharing among multiple equipment platforms and other functions. Such trend will expand the apparent damage information in archaeological research, preventive preservation and demonstrate dissemination of research and application value. Furthermore, it has the potential to support the digital twin construction of historic buildings with more comprehensive data.

CRediT authorship contribution statement

Hongchao Liu: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Ying Wu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Aiqun Li:** Supervision, Resources, Methodology, Conceptualization. **Yang Deng:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was funded by the National Natural Science Foundation of China under Grant No. 51978033 and No.52278290, the Pyramid Talent Training Project of Beijing University of Civil Engineering and Architecture (Grant No. JDYC20220830), and the BUCEA Doctor Graduate Scientific Research Ability Improvement Project (DG2022009).

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